MMVistaReason

- Open-data construction and capacity-aware multimodal post-training
- October 9 recommendation, anchored to arXiv v1
- A 4B/9B recipe spanning analytical and real-world visual reasoning

> 本 deck 固定阅读 v1。arXiv 页面存在 10 月 22 日 v2，本次不把新版本结论混入推荐时版本。

Paper v1 Project

Two reasoning groups define the pipeline

- Analytical tasks include science, mathematics, visual logic, and charts.
- Real-world tasks include perception, counting, spatial reasoning, and GUI grounding.
- Separate specialization addresses negative transfer under mixed-domain RL.

> 分组同时影响数据标签、RL expert 路由与后续蒸馏，并非仅用于最终报表。第 1、3 节。

Paper v1 Project

Cleaning preserves distinct questions per image

- SHA-256, pHash, and CLIP similarity detect redundant image/question pairs.
- The same image can retain multiple distinct questions.
- Qwen3.5-122B-A10B normalizes language, answerability, instructions, and answer forms.

> 一图多问不应按图片 ID 整体删除。语义清洗也可能改写任务，生产环境应保留原文和转换记录。第 3.1 节。

Paper v1 Project

Teacher cascades control supervision cost

- A 4B model generates four responses for initial difficulty filtering.
- Unresolved examples escalate through 9B, 27B, and 122B teachers.
- A verifier checks answer correctness and reasoning consistency before accepting trajectories.

> 级联路由把昂贵 teacher 的调用集中在前级未解样本上。它产生的是相对于当前模型族的难度。第 3.2 节。

Paper v1 Project

Answer-Likelihood Best-of-N

- Candidate traces first pass correctness and format checks.
- The teacher scores normalized final-answer log likelihood conditioned on each trace.
- Selection favors a valid trace that strongly supports the answer while controlling answer-length bias.

> 答案概率不是独立的人类推理质量证明。需要关注 verifier 偏差与 teacher 自洽偏好。第 3.2 节。

Paper v1 Project

Published supervision inventory

Subset	Teacher	Samples
MVR-SFT-355K	Qwen3.5-9B	354,886
MVR-SFT-124K	Qwen3.5-27B	124,010
MVR-SFT-49K	Qwen3.5-122B	48,913
Combined SFT	Cascade	527,809

> 表 1：总量约 1.04B response tokens。样本量与 token 成本都应记录，不能只比较条数。

Paper v1 Project

RL prompts follow each model’s frontier

- The corresponding SFT model produces eight rollouts per answer-verifiable prompt.
- Scale-specific pass-rate intervals keep inconsistently solvable prompts.
- Analytical and real-world subsets train separate GSPO experts from the SFT initialization.

> 4B 与 9B 不共享一个固定难度阈值。训练策略改变后，应重新测 pass rate，而不是永久贴上 hard 标签。第 3.2–3.3 节。

Paper v1 Project

Multi-teacher on-policy distillation

- The student generates trajectories and receives domain-routed expert supervision.
- Teacher top-32 distributions supervise the student's own rollout positions.
- The paper finds preferred KL direction varies with model scale.

> MOPD 融合 expert 的能力。不要把它描述为直接合并权重或简单拼接离线 teacher traces。第 3.3 与 4.4 节。

Paper v1 Project

Reported evaluation scope

Element	Reported configuration
Benchmarks	15 visual reasoning benchmarks
Decoding	Greedy, maximum 16,384 generated tokens
MVR-4B	72.8 overall average
MVR-9B	74.4 overall average

> 第 4 节报告值。总体平均由具体任务组成决定，不构成任意生产任务上的质量保证。

Paper v1 Project

Comparisons require matched supervision

- The authors compare Qwen3.5 Instruct and MMFineReason variants.
- Smaller students benefit more from selected supervision in the reported analyses.
- Mixed-domain RL can hurt individual benchmarks even when the overall average rises.

> 训练数据规模、token 长度、teacher 成本和评测集合均影响比较。70% fewer samples 是相对特定 MMFineReason 设置。

Paper v1 Project

Data platform architecture implications

- Inference: store source, cleaning recipe, verifier result, and teacher escalation path per sample.
- Proposed application: maintain pass-rate snapshots keyed by student checkpoint.
- Proposed application: audit domain mixtures and specialist regressions before distillation promotion.

> 把难度与模型 checkpoint 关联后才能复现 frontier filtering。代码项目存在，但本 deck 不声称已审计或跑通其训练实现。

Paper v1 Project

Reproduction and adoption gates

- Pin v1 datasets, model checkpoints, and recipe versions before selecting a subset.
- Compare SFT, mixed RL, separate experts, and MOPD under a common evaluation harness.
- Report per-domain regressions and total teacher/verifier tokens as well as average scores.

> 先在小规模上复核负迁移和蒸馏收益。版本变更应单列差异，不把 v2 当作 v1 的静默替换。

Paper v1 Project